

MONTH 1: Python Foundations & Programming Logic

Week 1 – Environment & Basics

- Install Python (Anaconda / venv)
- Setup VS Code / PyCharm
- Understand Python execution flow
- Variables & data types
- Type casting
- Input/output programs

Tasks

- Program to calculate simple & compound interest
 - Program to check prime, palindrome, armstrong numbers
-

Week 2 – Control Flow & Functions

- Conditional statements (`if` / `elif` / `else`)
- Loops (`for`, `while`)
- Loop control (`break`, `continue`, `pass`)
- Functions & return values
- Lambda functions

Tasks

- Number guessing game

- Menu-driven calculator
 - Pattern printing programs
-

Week 3 – Data Structures

- Lists & list methods
- Tuples & immutability
- Sets & set operations
- Dictionaries & key-value logic
- Nested data structures

Tasks

- Student marks management system
 - Frequency counter program
 - Dictionary-based contact book
-

Week 4 – File Handling & OOP

- File read/write (txt, csv)
- Exception handling
- Classes & objects
- Constructors
- Inheritance & encapsulation

Mini Project

- **CLI-based Library / Expense Management System**

Deliverables

- GitHub repo with:
 - `/month1/`
 - README.md
 - Well-commented code
-

MONTH 2: Data Analysis & Data Science

Week 5 – NumPy

- Arrays & indexing
- Array operations
- Broadcasting
- Statistical functions



Tasks

- Matrix operations
 - Data normalization
 - Random data simulation
-

Week 6 – Pandas

- Series & DataFrames
- Import CSV, Excel, JSON
- Filtering & sorting
- GroupBy operations

- Handling missing data

Tasks

- Sales data cleaning
 - Customer segmentation table
 - Monthly revenue report
-

Week 7 – Data Visualization

- Matplotlib basics
- Seaborn plots
- Line, bar, pie, histogram
- Correlation heatmaps

Tasks

- Visual EDA report
 - Business insights dashboard (static)
-

Week 8 – Statistics & EDA

- Mean, median, mode
- Variance & standard deviation
- Correlation & covariance
- Outlier detection

Mini Project (Real Dataset)

- Titanic Survival Analysis

- Data cleaning
- EDA
- Insights report

Deliverables

- Jupyter Notebook
 - PDF insights summary
 - GitHub push
-

MONTH 3: Machine Learning Fundamentals

Week 9 – ML Basics

- What is Machine Learning
- ML lifecycle
- Train-test split
- Feature scaling

Tasks

- Dataset preprocessing pipeline
-

Week 10 – Supervised Learning

- Linear Regression
- Multiple Regression

- Logistic Regression

Tasks

- House Price Prediction
 - Salary prediction model
-

Week 11 – Classification Models

- KNN
- Decision Tree
- Random Forest
- Naive Bayes

Tasks

- Customer churn prediction
 - Model comparison report
-

Week 12 – Unsupervised Learning

- K-Means clustering
- Elbow method
- PCA basics
- Model evaluation metrics

Mini Project

- **Customer Segmentation System**

Deliverables

- Model notebook
 - Evaluation metrics
 - README explaining approach
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MONTH 4: AI, Advanced ML & Capstone

Week 13 – Artificial Intelligence Basics

- AI vs ML vs DL
 - Rule-based systems
 - AI use cases
-

Week 14 – NLP

- Text preprocessing
- Tokenization
- Stopwords & stemming
- Sentiment analysis

Tasks

- Product review sentiment analyzer
-

Week 15 – Computer Vision (Intro)

- Image processing basics
- OpenCV fundamentals

- Face/object detection overview

Tasks

- Image classifier (basic)
-

Week 16 – Capstone + Deployment

- Dataset selection
- Problem statement
- Model building
- Final evaluation
- GitHub project structuring
- Resume & interview prep

Capstone Project (Any One)

- End-to-end ML prediction system
- Sales forecasting system
- Sentiment-based product analysis
- Recommendation system

GitHub Submission Rules (Mandatory)

- Public GitHub repository
- Naming: intern-name_python_ds_ai_ml
- Month-wise folders (month1, month2, etc.)
- README.md in every project
- Proper comments & clean code
- Capstone project mandatory for certificate